

Grade 6 Accelerated
Unit 10
Lesson 1 Homework

Name Answer Key
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The Customer Service Manager at a call center collected data on the number of minutes a customer was on hold before speaking to a representative. Her data is shown.

8	12	11	7	8	10	9	35	9	10	14
13	4	15	12	10	21	6	5	15	13	11

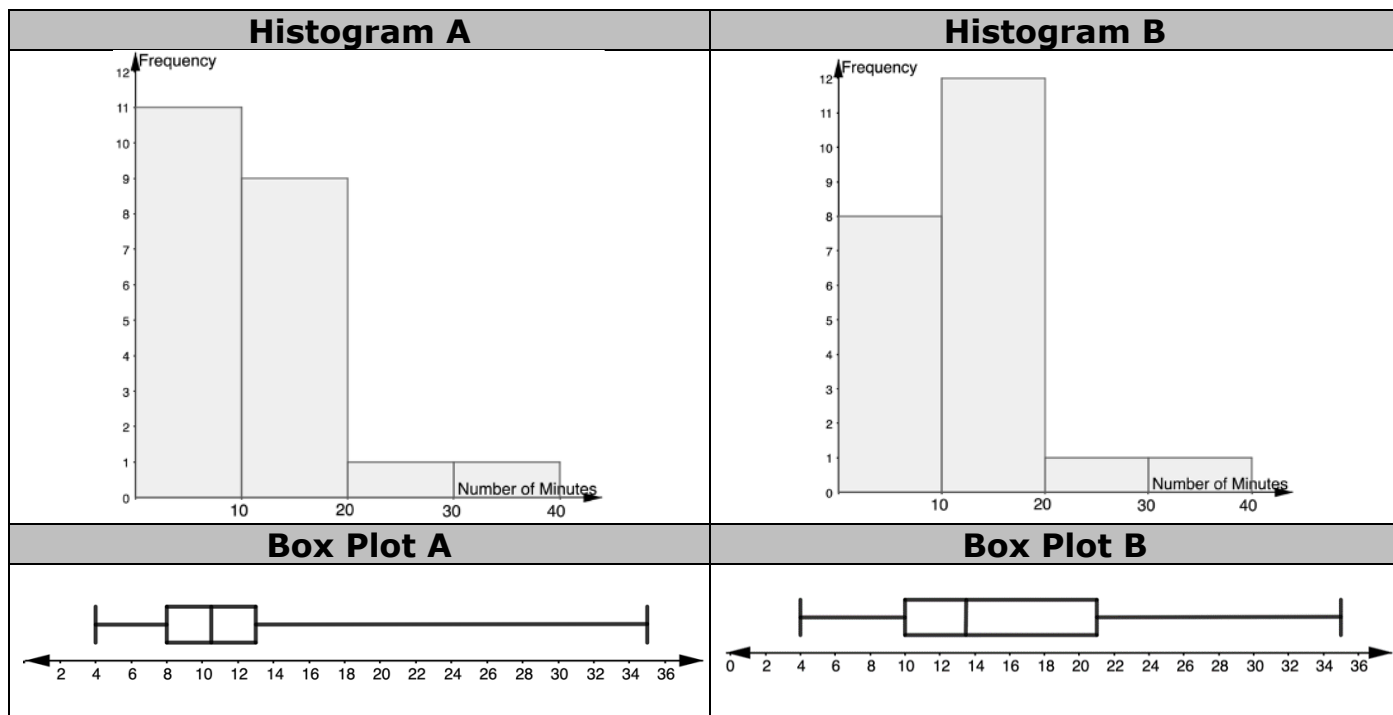
1. Create a frequency table of the data.

Number of Minutes	Frequency
0 – 10	8
10 – 20	12
20 – 30	1
30 – 40	1

2. Determine the five-number summary of the data set.

Min: 4
Q₁: 8
Median: 10.5
Q₃: 13
Max: 35

Two histograms and two box plots are shown.



3. Explain which histogram better represents the data.
Histogram B better matches my frequency table.
4. Explain which box plot better represents the data.
Box Plot A has the correct medians and quartiles.
5. Which graph, box plot or histogram, is better for displaying this data? Explain your reasoning.
Explanations may vary.
I think the histogram gives a better idea of the length of time customers are on hold most often.

Four data displays and four data sets are shown.

Data Display 1	Data Display 2
Data Display 3	Data Display 4
Data Set A	Data Set B
0, 0, 0, 1, 1, 1, 1, 2, 2, 2, 3, 10	0, 3, 4.5, 5, 6, 6.5, 7, 7, 7.5, 8, 9, 9.5
Data Set C	Data Set D
0, 0, 1, 1, 1, 2, 2, 2, 2, 3, 4, 6	0, 2, 4, 6, 6, 7, 7, 7, 7, 8, 8, 8

Match each Data Display to a Data Set.

6. Display 1: B
7. Display 2: C
8. Display 3: D
9. Display 4: A
10. Explain how you matched the data set that was represented by Data Display 3.
Explanations may vary.
The range for Display 3 is 0 to 8 and D is the only data set with the same range.

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Lesson 2 Homework

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Noelle kept track of the number of pages she read each day for the past 20 days. Her data are shown.

24, 35, 28, 27, 33, 36, 154, 28, 27, 25, 32, 25, 27, 31, 30, 28, 32, 25, 28, 30

1. One value in Noelle's data set is an outlier. Which value appears to be an outlier? Explain your reasoning.

154 because most values are in the 20's and 30's, so it's a more than 100 page difference.

2. Complete the statements.

The mean of Noelle's full data set is 35.25. $\frac{705}{20} = 35.25$

The mean of Noelle's data set with the outlier removed is 29. $\frac{551}{19} = 29$

3. Is there a significant difference between the mean of the full data set and that of the modified data set?

There is a 6.25 page difference which is somewhat significant.

4. Complete the statements.

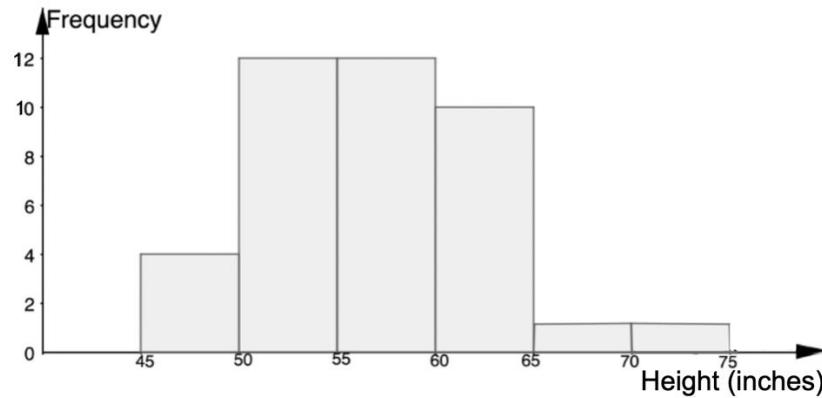
The IQR of Noelle's full data set is 5. $32 - 27$

The IQR of Noelle's data set with the outlier removed is 5. $32 - 27$

5. Is there a significant difference between the median of the full data set and that of the modified data set?

There was no difference; both are 28.

The histogram shows the heights, in inches, of students in a 6th grade class.



6. Describe the shape of the histogram.

Slightly right skewed

7. Explain which measure of center, mean or median, best represents this data.

Median because the graph is skewed

8. Explain which measure of variability, range or IQR, best represents this data.

IQR because the graph is skewed

The length of the songs on an album, in minutes, is shown.

3.5, 3.5, 3.75, 4, 4, 4, 4.25, 4.5, 5, 5, 5.5, 5.75

9. Explain which measure of center, mean or median, best represents this data.

The data appears to be symmetrical (no outliers); the mean better represents the data.

10. Explain which measure of variability, range or IQR, best represents this data.

Range because of the symmetrical data

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Lesson 3 Homework

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1. Fill in the blanks with the terms provided.

mean	range	median	interquartile range (IQR)
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The _____ is the middle of an ordered list of values. It is a measure of center, or central tendency.

The _____ is the arithmetic average of a set of numbers. It is a measure of center, or a central tendency.

The _____ is the difference between the highest data value and the lowest data value in a data set. It is a measure of variation.

A measure of variation in a set of numerical data, the _____ is the distance between the first and third quartiles of the data set.

The data for the number of emails a teacher gets on school days, Monday-Friday, was recorded for two weeks. Use the data shown for questions 2–8.

59	11	65	53	70
60	57	74	67	56

2. Determine the mean of the data set.

$57210=57.2$ The mean is approximately 57 emails.

3. Determine the median of the data set.

11 53 56 57 59 60 65 67 70 74

$59+602=59.5$ The median is 59.5 e-mails.

4. What is the range of the data set?

$74-11=63$ The range is 63 e-mails.

5. Which measures of center can be determined from a histogram?

No measures of center but the measure of variation, the range, can be determined from a histogram.

6. The presence of an outlier does not have a significant effect on the IQR. True or False

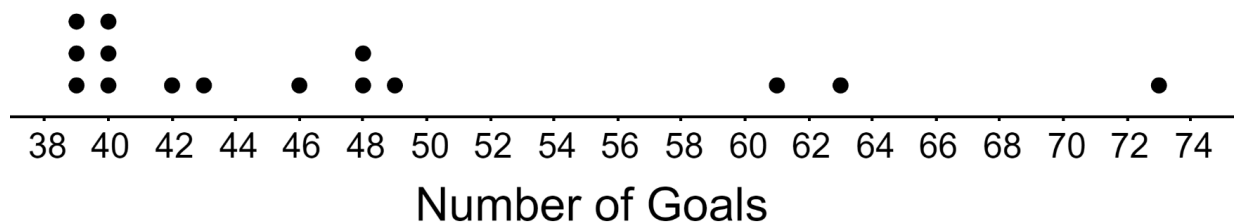
True

7. The presence of an outlier does have significant effect on the range. True or False

True

Below is a line plot of the number of goals scored by the top 15 women goal scorers during the 2019-2020 NCAA Women's Division 1 soccer season.

2019-2020 NCAA Women's Division 1 Soccer Top Scorers



8. Which measure of the center, the mean or median, would be used summarize this data?

Median because of outliers

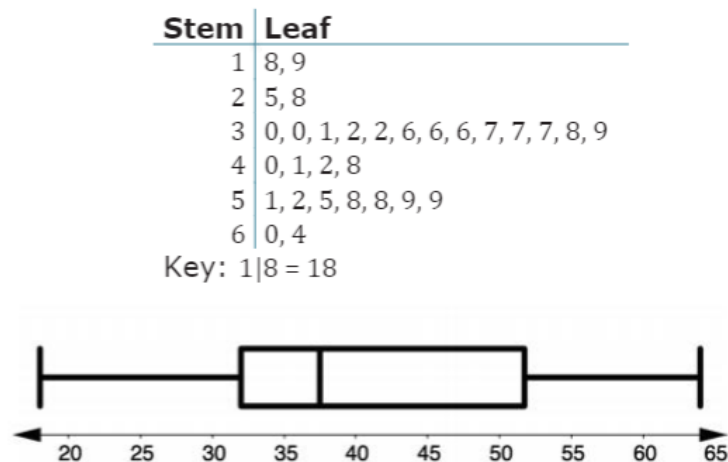
9. Explain whether the range or the interquartile range should be used to summarize the spread of the data.

IQR because of outliers

10. What is the range of the data?

$$73 - 39 = 34$$

Mr. Ramirez is trying to determine the average score of his student's quiz grades on a recent quiz. He wrote down a summary of the student scores and displayed them in the stem-and-leaf plot and box plot.



1. Which measure of center, the mean or median, best summarizes the quiz scores of the class?

Answers may vary.

Because there aren't any outliers, the mean would best summarize the scores.

2. Find the value of the measure you chose in question 1 if possible.

Mean: $\frac{1,228}{30} \approx 40.9$

Median: 37.5

3. Describe the variation of quiz scores.

Range is $64 - 18 = 46$.

The IQR is $52 - 32 = 20$.

4. Does the range or the interquartile range better represent the variation in quiz scores?

Answers may vary.

Because there aren't any outliers, the range best represents the variation.

Jackie is planning on buying a new car. She has a budget of \$45,000. The prices of new cars in her area are shown below, in thousands of dollars.

23, 52, 42, 38, 56, 70, 40

5. Which measure of center should Jackie use to summarize the prices of the new cars in her area, the mean or the median? Explain your reasoning.
Answers may vary.
Because of the outliers (23 and 70), she should use the median.
6. Which measure of variation should Jackie use to summarize the prices of the new cars in her area, the IQR or the range? Explain your reasoning.
Answers may vary.
IQR because of the outliers.

Jorge likes to listen to music on his phone. The data set below shows the times of the songs on his playlist.

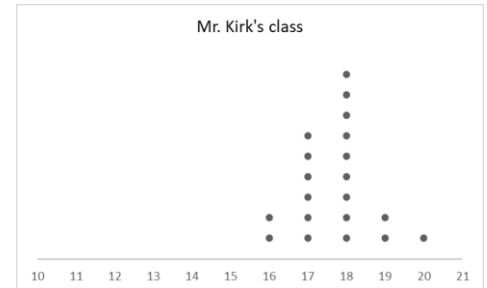
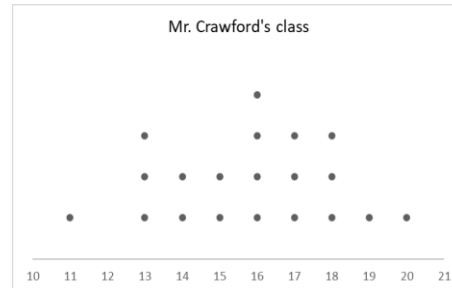
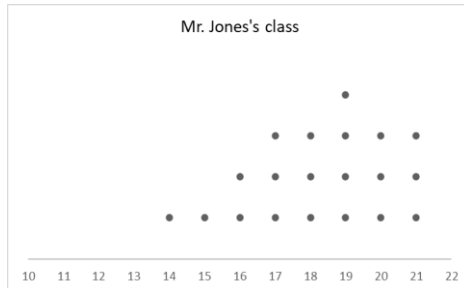
3.25, 1.58, 2.22, 4.03, 3.57, 2.41, 7.46, 3.13, 4.01, 2.59

7. Are there any outliers in the data set? If so, which one?
Explanations may vary.
7.46 is an outlier because it is so much larger than most of the other songs.
8. What measure of center best describes the typical song length on Jorge's playlist? Explain your reasoning.
Explanations may vary.
Median because of the presence of the outlier.
9. What measure of variation best describes the spread of the data distribution? Explain your reasoning.
Explanations may vary.
IQR because of the outlier.
10. Jorge says the IQR is the most accurate depiction of the spread of his song lengths. Do you agree or disagree with Jorge? Explain your reasoning.
Explanations may vary.
I agree because the outlier will affect the range, so the IQR would be more accurate.

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Lesson 5 Homework

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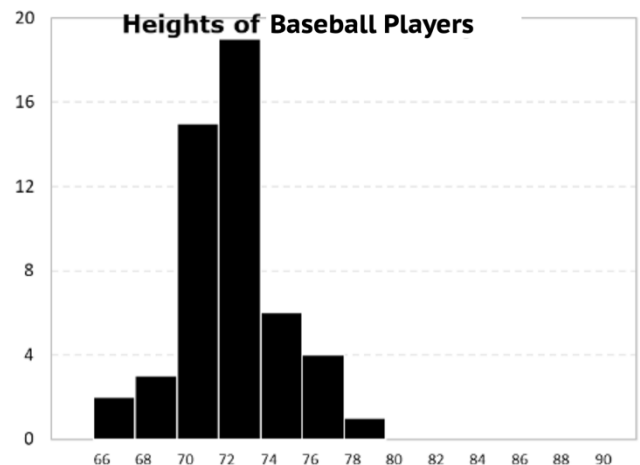
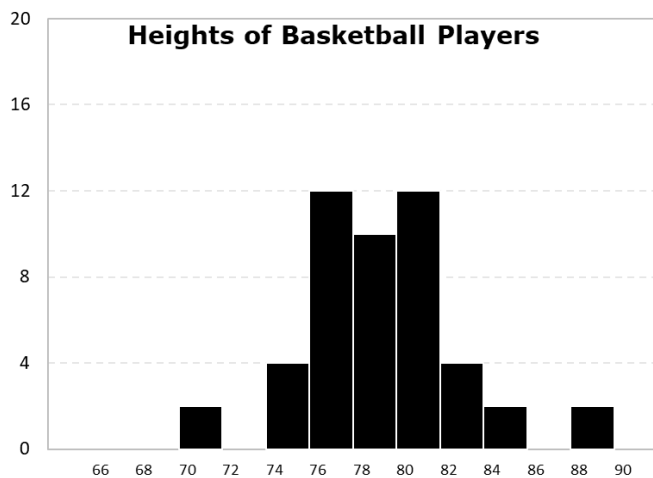
The line plots below show the number of hours the students in three different classes spend watching videos on the internet each week.



Time (in hours)

- Which line plot shown has a center of data that is greater than the center of data for Mr. Kirk's class?
Mr. Jones's class
- The value of the largest mean number of hours of all three classes is 18.25.
- The value of the largest median number of hours of all three classes is 18.5.
- Which of the line plots shown has the most variability in the times spent watching videos on the internet? **Mr. Crawford's class**
Jones: $21 - 14 = 7$
Crawford: $20 - 11 = 9$
Kirk: $20 - 16 = 4$
- Based on the data from the line plots shown, which class spends the most time watching videos on the internet each week?
Explanation may vary.
Because his class has the highest mean and median, Mr. Jones's class spends the most time watching videos on the internet each week.

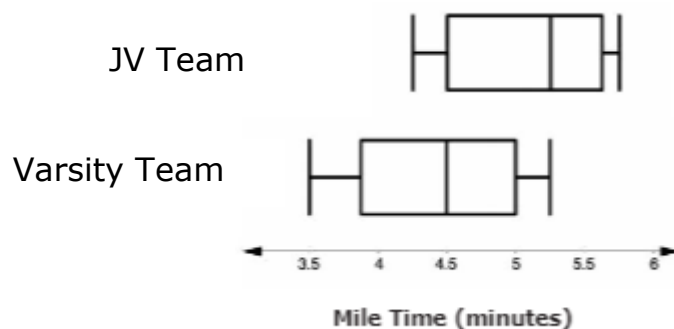
The two histograms show height distributions of 50 male professional basketball players and 50 male professional baseball players, in inches (in).



Height (in.)

6. The group with the larger center of data is basketball players.
7. The group with the larger range of heights is basketball players.
8. In one to two sentences use the words shape, center, and variability to describe the differences between the two histograms. **Answers may vary.**
Baseball: slightly left skewed with majority between 70 – 74
Basketball: has outliers which creates a greater range and is asymmetric

The box plots shown below are the mile times for a JV and Varsity volleyball team.



9. Calculate and describe how the centers compare between the two teams.
JV median: 5.25 min.
Varsity median: 4.5 min.
Varsity has a lower median mile time than JV.
10. Which team has the bigger variability? Explain your reasoning.
Varsity's Range (1.75) and IQR (≈ 1.25) are larger than JV's range (1.5) and IQR (≈ 1.1)

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Lesson 6 Homework

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A randomly selected group of male and female students were selected to have their heights measured at their middle school. The heights, in inches, of the 15 randomly selected male and female students are shown below.

Male students: 56, 50, 49, 74, 46, 52, 61, 49, 53, 56, 60, 58, 47, 51, 59

Female students: 40, 56, 50, 60, 66, 65, 55, 54, 52, 50, 49, 62, 61, 59, 64

Complete table by using the heights of the male and female students shown above.

Measure	Male Students	Female Students
1. Mean	$\frac{821}{15} = 54.\overline{73}$	$\frac{843}{15} = 56.2$
2. Median	53	56
3. Range	$74 - 46 = 28$	$66 - 40 = 26$
4. IQR	$59 - 49 = 10$	$62 - 50 = 12$
5. Potential Outliers	74	40

6. Explain which measure you would use to describe the data sets above.

Because both male and female students have outliers, the median and IQR would give better information about the data.

7. Which of the data sets have more variability?

The IQR is larger in the heights of female students so that data has more variability.

A fast food restaurant was trying to determine which way customers could get their food the fastest, inside the lobby or in the drive thru. The restaurant paid 9 mystery shoppers to time how quickly they received their food in the lobby and in the drive thru. The times, in minutes are shown below.

	1	2	Q ₁ 2.5	3	4	Med 5	6	7	Q ₃ 7.5	8	15
Lobby	2	15		7	5	8	3	4	6		1
Drive Thru	5	3		6	5	4	6	2	7		9
	2	3	Q ₁ 3.5	4	5	Med 5	6	6	Q ₃ 6.5	7	9

8. What are the mean and median times for getting food in the lobby and in the drive thru?

	Lobby	Drive Thru
Median	5	5
Mean	$\frac{51}{9} = 5.\bar{6} = 5\frac{2}{3}$	$\frac{47}{9} = 5.\bar{2}$ or $5\frac{2}{9}$

9. Which method of getting food has the greatest IQR? Which method has the biggest range?

	Lobby	Drive Thru
IQR	$7.5 - 2.5 = 5$	$6.5 - 3.5 = 3$
Range	$15 - 1 = 14$	$9 - 2 = 7$

The lobby has the higher IQR and range.

10. Based on the times shown above, use the most appropriate measures of center and variability to determine which method you would choose to get your food from this restaurant.

The drive thru has a shorter average wait time with a smaller variability, so the drive thru is the quicker choice.

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Lesson 7 Homework

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A group of students challenged each other to see how long it took them to finish the same math problem. The students want to see who can solve the problem correctly the fastest. The times of the students are listed in seconds is represented in the stem and leaf plot.

Stem	Leaf
0	80
1	02 15 45 59 60
2	04 32 45
3	33

Key: 1|02 = 102

- Find the mean and median of the data set.
Median: 159.5 seconds
Mean: $\frac{1775}{10} = 177.5$ seconds
- The range of the data set is **$333 - 80 = 253$** .
- The IQR of the data set is **$232 - 115 = 117$** .

A different group of students in the same class timed themselves to see how long it took them to solve the same math problem and recorded their time in seconds.

{134, 122, 111, 90, 101, 103, 143, 60, 45, 100}

- Find the mean and median of the data set.
Median: 102 seconds
Mean: $\frac{1009}{10} = 100.9$ seconds
- The range of the data set is **$143 - 45 = 98$** .
- The IQR of the data set is **$112 - 90 = 22$** .

For questions 7–10, determine whether each statement is true or false. Explain your reasoning.

7. The first group of students had a lower median than the second group of students.

False. The median of the first group was 159.5 seconds which is higher than the second group's, 102 seconds.

8. The center of group 1 is similar to the center of group 2.

False. The center measures for group 1 (177.5, 159.5) were both larger than the center measures for group 2 (100.9, 102).

9. The second group of students has a larger range than the first group of students.

False. The range of the second group, 98, is much smaller than the range of the first group, 253.

10. The first group of students is more consistent with their times than the second group.

False. Explanations may vary.

Group 2's mean and median were close in value, and group 2's range and IQR were smaller. Therefore, group 2 was more consistent with their times.

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Lesson 8 Homework

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Scenario 1: Imagine a market researcher surveys 65 people on their coffee-drinking habits. He wants to know whether people in the local region are willing to switch their regular drink to something new.

1. In this scenario, what is the sample?

65 people surveyed

2. In this scenario, what is the population?

People in the local region

Scenario 2: Jasmine wants to collect data on the favorite dessert by young people in Jacksonville, FL. She decides to go outside a local store and ask every person who comes into the store during the morning and afternoon about their dessert preferences.

3. What is the population?

Young people in Jacksonville, FL

4. What is the sample?

People who go to the local store during the morning and afternoon

5. Explain why a sample chosen in this way may not be representative of the population.

People who go to the store would include all ages, not just young people.

Scenario 3: John is going to sell lemonade outside his house and save money in order to buy a new laptop. He wants to predict his sales and estimate how many days he must sell lemonade. He is considering three different methods for surveying a sample. Describe the benefits and problems with each of the possible survey methods.

Answers may vary.

6. Ask everyone in his class if they like lemonade and how much will they drink in a day.

A benefit is that his classmates have a variety of wants. A problem is that his classmates are not a representative sample of people he is selling lemonade to outside his house.

7. Ask all people younger than 15 around his house if they like lemonade and how much will they drink in a day.

A benefit is that there will be a variety of wants.
A problem is this is not a representative of the population of all ages buying lemonade.

8. Ask all neighbors around his house if they like lemonade and how much will they drink in a day.

A benefit is that this is representative of the population of people buying lemonade.
A problem is that it does not incorporate people passing by who are not neighbors.

9. Explain which of the methods from questions 6, 7, and 8 would be the most likely to produce a sample that is representative of the population.

Asking all neighbors would be the most likely to produce a representative sample.

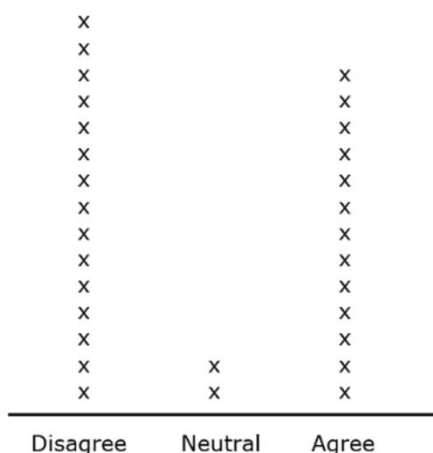
10. Determine a better way to select a sample for this study.

Ask every 5th person who passes by his house for one week.

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Lesson 9 Homework

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A middle school in Warner Robins, GA is considering requiring students to wear uniforms for the upcoming school year. The decision was based on proposals by the school advisory committee and PTSA. Parents, teachers, and staff overwhelmingly agreed that wearing uniforms will have a positive effect on the school. The school's principal conducted a random survey of 30 students to gauge student opinions on the potential mandatory uniform policy. The results are shown on the line plot below. Use the data to answer questions 1-5.



1. What fraction of students surveyed were neutral on the potential mandatory school uniform policy?

$$\frac{2}{30} \approx 6.67\%$$

2. A study conducted by a research group in Nevada indicated that 54% of middle school students agreed with a mandatory uniform policy. Explain how this compares to the students at the school in Warner Robins that agree with a mandatory uniform policy.

The 54% is higher than the 43% sample at the school in Warner Robins.

3. There are approximately 3,512 middle students in Warner Robins, Georgia public schools. Predict the number of students in Warner Robins that would disagree with a mandatory uniform policy based on the data from the line plot.

$$50\% \text{ of } 3,512 = 0.5 (3,512) = 1,756 \text{ students would disagree.}$$

4. There are approximately 3,512 middle students in Warner Robins, Georgia public schools. Predict the number of students in Warner Robins that would be neutral about a mandatory uniform policy based on the data from the line plot.

$$6.67\% \text{ of } 3,512 = .0667 (3,512) \approx 234 \text{ students would be neutral.}$$

5. Explain at least *two* factors that could contribute to the differences between your estimates and the actual number of students who could have voted about a mandatory uniform policy.

Answers may vary.

- 1) A sample of 30 students is very small for a population of 3512 students.
- 2) The sample did not indicate how the random survey was given, which could create a biased sample.

A guidance counselor at Top Performance High School in Washington, DC wants to analyze scores on the SAT to increase the average scores for seniors. The counselor gathered the SAT scores for all seniors at the school. The average SAT score in the US is 1051.

The counselor picked a random sample of 200 senior scores and recorded the information in the table below. Use this data to answer questions 6-10.

Students that scored under 1051	Students that scored 1051 and higher.
86	114

6. What is the proportion of students that scored 1051 or greater on the SAT at Top Performance High School?

$$\frac{114}{200}$$

7. Estimate the probability of a senior in the random sample scoring 1051 or greater on the SAT? Explain your method.

$$\frac{114}{200} = 0.57 = 57\% \quad \text{divide number of students scoring 1051 or greater by total in sample.}$$

8. There are 638 seniors at Top Performance High School. Using equivalent fractions, how many seniors are expected to score 1051 or greater on the SAT (based on the sample data)?

$$\frac{114}{200} \times \frac{3.19}{3.19} \approx \frac{364}{638} \quad 364 \text{ students}$$

9. If you use the percent method to answer question 8, what similarities and differences do you notice?

$$0.57 (638) \approx 364 \quad \text{Similarity: same answer with different method.}$$

Difference: multiply percent by 364.

10. The guidance counselor estimates that 400 seniors will score 1051 or greater on the SAT sample data sample data. Explain whether this would be considered an unreasonable based on the sample data.

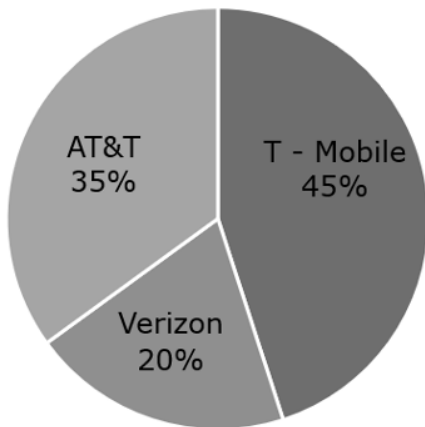
400 students are considerably more than the 364 estimated to score 1051 or greater, so this could be considered unreasonable.

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550 Students in the 8th grade were polled to see what cell phone carrier was the most common in a local middle school. The middle school has a total population of 2475 students. The circle graph below shows the percentage of students that have the three major cell phone carriers: AT&T, T-Mobile and Verizon. Use the circle graph to answer questions 1-3.

Student Cell Phone Carriers



1. To predict how many students with Verizon as a carrier, the student council president wants to use the scale factor method. What scale factor should she use to make this prediction?

$$\frac{2,475}{550} = 4.5$$

2. Explain whether the equivalent factor or percent method be used to predict the number of students with AT&T as a carrier should the equivalent factor or percent method be used?

Answers may vary. Percent method because they're given in the circle graph.

3. Predict the number of students that have T-Mobile as a cell phone carrier.

$$0.45 (2,475) = 1,113.75 \approx 1,114 \text{ students have T-Mobile.}$$

Tyrone wants to know about the color distribution of Skittles in his candy jar. Without looking he picked out one Skittle at a time, recorded it, and put them back into the jar. Tyrone picked 10 green, 6 purple, 12 red, 8 yellow, 4 orange after 40 picks. A total of 220 Skittles are in the jar. Use Tyrone's results to answer questions 4-5.

4. What fraction of Skittles are green, purple, red, yellow and orange?

$$\text{Green: } \frac{10}{40} = \frac{1}{4}$$

$$\text{Red: } \frac{12}{40} = \frac{3}{10}$$

$$\text{Orange: } \frac{4}{40} = \frac{1}{10}$$

$$\text{Purple: } \frac{6}{40} = \frac{3}{20}$$

$$\text{Yellow: } \frac{8}{40} = \frac{1}{5}$$

5. Assuming his sample is a true representation of Skittle distribution, how can Tyrone predict the estimated number of Skittles in the jar that are red?

Answers may vary. Multiply the fraction of red Skittles by the total number of skittles in the jar.

The cheerleading team at a local school cannot attend all of the games for all sports teams in the month of February due to conflicting schedules. The Athletic Director, Mrs. James, wants to prioritize the games the team will cheer at to boost morale of the teams. Mrs. James asks 150 students to pick the sport the cheerleading team should cheer. The results were recorded in the table below. The school has a total of 1150 students. Use the results of the sample to answer questions 6-10.

Sport	Number of Times Chosen	Ratio of Students that Picked Each Sport
Baseball	25	$\frac{25}{150} = \frac{1}{6}$
Basketball	55	$\frac{55}{150} = \frac{11}{30}$
Volleyball	40	$\frac{40}{150} = \frac{4}{15}$
Soccer	30	$\frac{30}{150} = \frac{1}{5}$

6. Find the ratio of students in the sample that picked each sport and complete the table.

7. Predict the number of students in the school that want the cheerleading team to perform at the baseball games (based on sample results) using the percent method.

$$\frac{1}{6} (1,150) = 191.\bar{6} \approx 192 \text{ students}$$

8. Explain which is the best method to predict the number of students in the school that want the cheerleading team to perform at the soccer games (based on sample results).

Answers may vary. Percent strategy because we have the fraction form of the percent.

9. Mrs. James uses the scale factor method to predict the number of students in the school that want the cheerleading team to cheer at basketball games. She uses a scale factor of 0.13. Did Mrs. James use the right scale factor? Explain.

$$\text{No because } \frac{\text{population}}{\text{sample}} = \frac{1150}{150} = 7.\bar{6}$$

10. The cheerleading team prefers to cheer at indoor sports. Predict the amount of students that prefer the cheerleading team to perform at the indoor sports.

$$\frac{55+40}{150} = \frac{95}{150} (1,150) = 728.\bar{3} \approx 728 \text{ students}$$